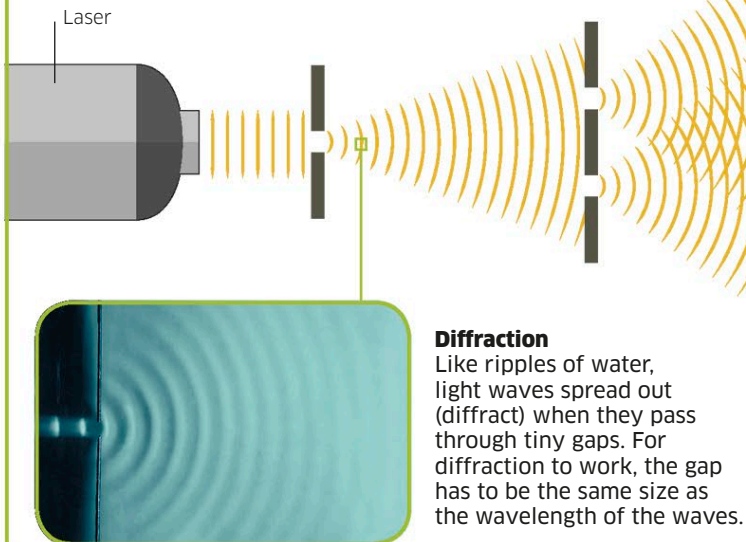


Diffraction and interference

Light waves spread out when they pass through tiny gaps or holes. The smaller the gap, the more spreading (diffraction) that occurs. When two or more waves meet, they add together or cancel each other out, forming bigger or smaller waves. This is known as interference.

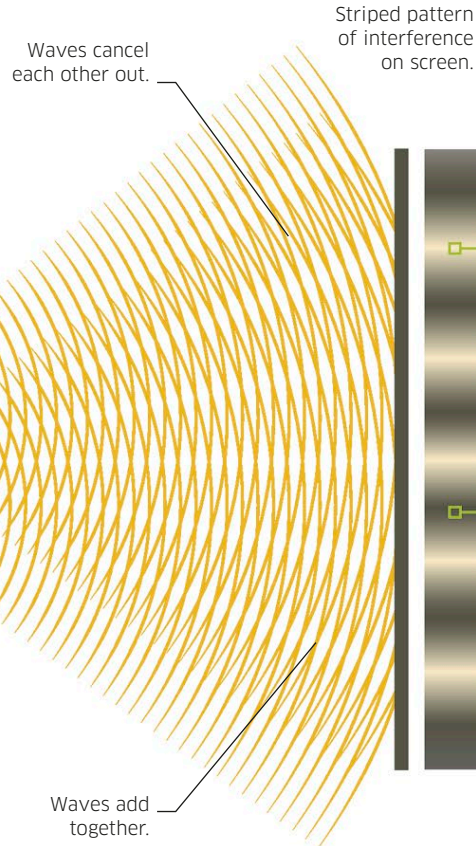
Double-slit experiment

To prove that light behaves as a wave, not a particle, in 1801 English scientist Thomas Young shone light through slits to demonstrate that light waves diffract and interfere like waves in water.



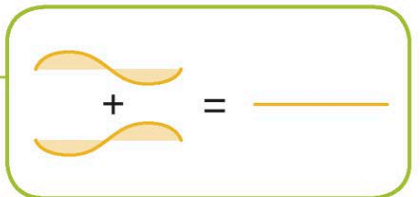
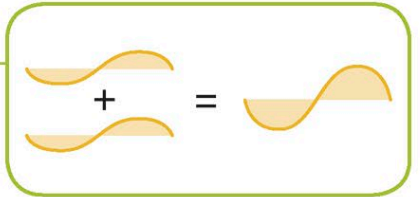
Diffraction

Like ripples of water, light waves spread out (diffract) when they pass through tiny gaps. For diffraction to work, the gap has to be the same size as the wavelength of the waves.



Constructive interference

When two waves of the same length and height (amplitude) overlap in phase, they add together to make a new wave that has twice the height, making a light twice as bright.



Destructive interference

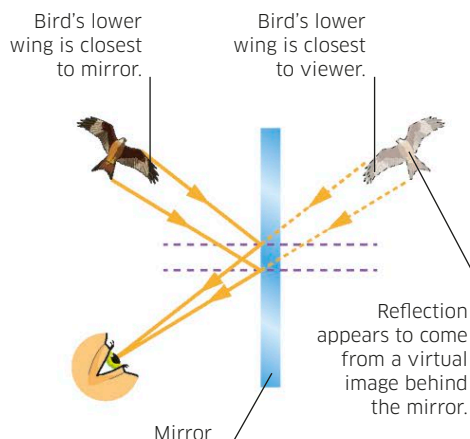
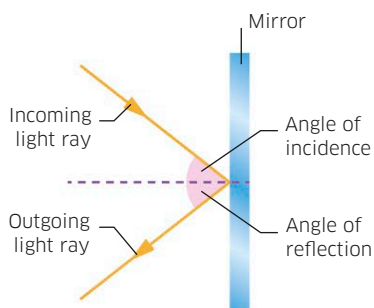
When two identical waves add together, but are out of phase, they cancel each other out. The wave they make has zero amplitude, making darkness.

Reflection

Light rays bounce off a smooth surface, such as a mirror, in a single beam. This is called specular reflection. If the surface is rough, the rays bounce off randomly in different directions. This is called diffuse reflection.

The law of reflection

A light ray beamed at a mirror bounces off again at exactly the same angle, or, in more scientific terms, the angle of incidence is equal to the angle of reflection.



Reverse images

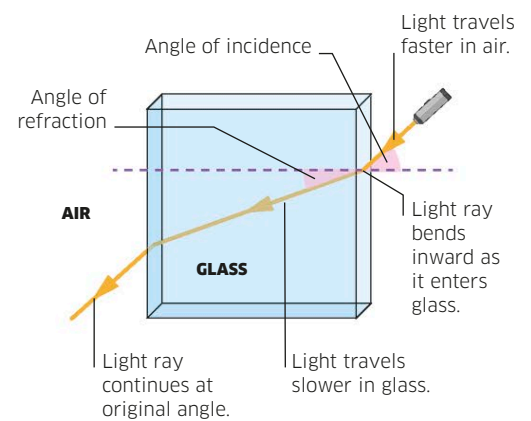
Mirrors don't reverse things left to right—writing looks reversed because you've turned it around. What mirrors do is to reverse things back to front along an axis at right angles to the mirror.

Refraction

Light rays travel more slowly in more dense substances such as water and glass than in air. The change in speed causes light to bend (refract) as it passes from air to glass or water and back. How much a material refracts light is known as its refractive index.

Bending light

Light rays slow down and bend as they pass from air to glass, and speed up and bend outward as they pass from glass to air. The refractive index of air is 1. For glass, it is around 1.60, depending on the quality of the glass, whereas for diamond—which is harder and denser—it is 2.40.



Real and apparent depth

Refraction makes an object in water appear nearer the surface. Because our brains assume that light rays travel in a straight line, rather than bending, we see the object in the water higher up than it really is. For a person under water, the reverse applies: an object on land appears higher up than it is.

